

```
/* =====  
ORACLE HR DATABASE LAB SCRIPT  
Topics: DDL, DML, DQL, DTL/TCL, DCL  
===== */
```

```
/* =====  
SECTION 1: DDL - DATA DEFINITION LANGUAGE  
Create HR tables  
===== */
```

```
DESC departments;  
DROP TABLE departments CASCADE CONSTRAINTS;
```

```
CREATE TABLE departments (  
    department_id NUMBER PRIMARY KEY,  
    department_name VARCHAR2(100) NOT NULL,  
    location VARCHAR2(100)  
);
```

```
DESC jobs;  
DROP TABLE jobs CASCADE CONSTRAINTS;
```

```
CREATE TABLE jobs (  
    job_id NUMBER PRIMARY KEY,  
    job_title VARCHAR2(100) NOT NULL,  
    min_salary NUMBER(10,2),  
    max_salary NUMBER(10,2)  
);
```

```
DESC employees;  
DROP TABLE employees CASCADE CONSTRAINTS;
```

```
CREATE TABLE employees (  
    employee_id NUMBER PRIMARY KEY,  
    first_name VARCHAR2(50) NOT NULL,  
    last_name VARCHAR2(50) NOT NULL,
```

```
email VARCHAR2(100) UNIQUE,  
phone VARCHAR2(30),  
hire_date DATE DEFAULT SYSDATE,  
salary NUMBER(10,2) CHECK (salary > 0),  
department_id NUMBER,  
job_id NUMBER,  
manager_id NUMBER,  
CONSTRAINT fk_emp_dept  
    FOREIGN KEY (department_id)  
    REFERENCES departments(department_id),  
CONSTRAINT fk_emp_job  
    FOREIGN KEY (job_id)  
    REFERENCES jobs(job_id),  
CONSTRAINT fk_emp_manager  
    FOREIGN KEY (manager_id)  
    REFERENCES employees(employee_id)  
);
```

DESC attendance;

DROP TABLE attendance CASCADE CONSTRAINTS;

```
CREATE TABLE attendance (  
    attendance_id NUMBER PRIMARY KEY,  
    employee_id NUMBER NOT NULL,  
    attendance_date DATE DEFAULT SYSDATE,  
    check_in TIMESTAMP,  
    check_out TIMESTAMP,  
    status VARCHAR2(20) CHECK (status IN ('Present', 'Absent', 'Leave', 'Late')),  
    CONSTRAINT fk_att_emp  
        FOREIGN KEY (employee_id)  
        REFERENCES employees(employee_id)  
);
```

DESC payroll;

```
CREATE TABLE payroll (  

```

```
payroll_id NUMBER PRIMARY KEY,  
employee_id NUMBER NOT NULL,  
salary_month VARCHAR2(20),  
basic_salary NUMBER(10,2),  
allowance NUMBER(10,2),  
deduction NUMBER(10,2),  
net_salary NUMBER(10,2),  
CONSTRAINT fk_payroll_emp  
    FOREIGN KEY (employee_id)  
    REFERENCES employees(employee_id)  
);
```

```
/* Create sequences */
```

```
CREATE SEQUENCE seq_department START WITH 1 INCREMENT BY 1;  
CREATE SEQUENCE seq_job START WITH 1 INCREMENT BY 1;  
CREATE SEQUENCE seq_employee START WITH 1 INCREMENT BY 1;  
CREATE SEQUENCE seq_attendance START WITH 1 INCREMENT BY 1;  
CREATE SEQUENCE seq_payroll START WITH 1 INCREMENT BY 1;
```

```
/* Create indexes */
```

```
CREATE INDEX idx_emp_department ON employees(department_id);  
CREATE INDEX idx_emp_job ON employees(job_id);  
CREATE INDEX idx_attendance_date ON attendance(attendance_date);
```

```
/* Add one column */
```

```
ALTER TABLE employees  
ADD employment_status VARCHAR2(20) DEFAULT 'Active';
```

```
/* =====  
SECTION 2: DML - DATA MANIPULATION LANGUAGE  
Insert, Update, Delete  
===== */
```

```
/* Insert departments */
```

```
INSERT INTO departments VALUES  
(seq_department.NEXTVAL, 'Human Resources', 'Yangon');
```

```
INSERT INTO departments VALUES  
(seq_department.NEXTVAL, 'Finance', 'Mandalay');
```

```
INSERT INTO departments VALUES  
(seq_department.NEXTVAL, 'IT', 'Naypyidaw');
```

```
INSERT INTO departments VALUES  
(seq_department.NEXTVAL, 'Sales', 'Yangon');
```

```
/* Insert jobs */
```

```
INSERT INTO jobs VALUES  
(seq_job.NEXTVAL, 'HR Officer', 400000, 900000);
```

```
INSERT INTO jobs VALUES  
(seq_job.NEXTVAL, 'Accountant', 500000, 1200000);
```

```
INSERT INTO jobs VALUES  
(seq_job.NEXTVAL, 'Software Engineer', 700000, 2000000);
```

```
INSERT INTO jobs VALUES  
(seq_job.NEXTVAL, 'Sales Executive', 400000, 1500000);
```

```
INSERT INTO jobs VALUES  
(seq_job.NEXTVAL, 'IT Manager', 1500000, 3500000);
```

```
/* Insert employees */
```

```
INSERT INTO employees (  
    employee_id, first_name, last_name, email, phone,  
    hire_date, salary, department_id, job_id, manager_id  
)  
VALUES (  
    seq_employee.NEXTVAL, 'Aung', 'Aung', 'aung.aung@company.com',  
    '09111111111', DATE '2022-01-10', 2500000, 3, 5, NULL  
);
```

```
INSERT INTO employees (  
    employee_id, first_name, last_name, email, phone,  
    hire_date, salary, department_id, job_id, manager_id  
)  
VALUES (  
    seq_employee.NEXTVAL, 'Su', 'Su', 'su.su@company.com',  
    '09222222222', DATE '2023-03-15', 800000, 1, 1, 1  
);
```

```
INSERT INTO employees (  
    employee_id, first_name, last_name, email, phone,  
    hire_date, salary, department_id, job_id, manager_id  
)  
VALUES (  
    seq_employee.NEXTVAL, 'Kyaw', 'Kyaw', 'kyaw.kyaw@company.com',  
    '09333333333', DATE '2023-06-20', 950000, 2, 2, 1  
);
```

```
INSERT INTO employees (  
    employee_id, first_name, last_name, email, phone,  
    hire_date, salary, department_id, job_id, manager_id  
)  
VALUES (  
    seq_employee.NEXTVAL, 'Hla', 'Hla', 'hla.hla@company.com',  
    '09444444444', DATE '2024-02-01', 1200000, 3, 3, 1  
);
```

```
INSERT INTO employees (  
    employee_id, first_name, last_name, email, phone,  
    hire_date, salary, department_id, job_id, manager_id  
)  
VALUES (  
    seq_employee.NEXTVAL, 'Mya', 'Mya', 'mya.mya@company.com',  
    '0955555555', DATE '2024-05-05', 700000, 4, 4, 1  
);
```

```
/* Insert attendance */
```

```
INSERT INTO attendance VALUES  
(seq_attendance.NEXTVAL, 1, DATE '2026-05-01',  
    TIMESTAMP '2026-05-01 08:55:00',  
    TIMESTAMP '2026-05-01 17:30:00',  
    'Present');
```

```
INSERT INTO attendance VALUES  
(seq_attendance.NEXTVAL, 2, DATE '2026-05-01',  
    TIMESTAMP '2026-05-01 09:20:00',  
    TIMESTAMP '2026-05-01 17:00:00',  
    'Late');
```

```
INSERT INTO attendance VALUES  
(seq_attendance.NEXTVAL, 3, DATE '2026-05-01',  
    TIMESTAMP '2026-05-01 08:50:00',  
    TIMESTAMP '2026-05-01 17:10:00',  
    'Present');
```

```
INSERT INTO attendance VALUES  
(seq_attendance.NEXTVAL, 4, DATE '2026-05-01',  
    NULL,  
    NULL,  
    'Leave');
```

```
INSERT INTO attendance VALUES  
(seq_attendance.NEXTVAL, 5, DATE '2026-05-01',
```

```
TIMESTAMP '2026-05-01 09:00:00',  
TIMESTAMP '2026-05-01 17:00:00',  
'Present');
```

```
/* Insert payroll */
```

```
INSERT INTO payroll VALUES  
(seq_payroll.NEXTVAL, 1, 'May-2026', 2500000, 300000, 100000, 2700000);
```

```
INSERT INTO payroll VALUES  
(seq_payroll.NEXTVAL, 2, 'May-2026', 800000, 100000, 30000, 870000);
```

```
INSERT INTO payroll VALUES  
(seq_payroll.NEXTVAL, 3, 'May-2026', 950000, 150000, 50000, 1050000);
```

```
INSERT INTO payroll VALUES  
(seq_payroll.NEXTVAL, 4, 'May-2026', 1200000, 200000, 80000, 1320000);
```

```
INSERT INTO payroll VALUES  
(seq_payroll.NEXTVAL, 5, 'May-2026', 700000, 80000, 20000, 760000);
```

```
SQL> SELECT employee_ID, salary, department_id from employees  
WHERE department_id = 3;  
2  
EMPLOYEE_ID      SALARY DEPARTMENT_ID  
-----  
1      2500000      3  
4      1200000      3
```

```
/* Update employee salary */
```

```
UPDATE employees  
SET salary = salary + 100000  
WHERE department_id = 3;
```

/* Delete inactive attendance example */

```
SQL> SELECT attendance_id, employee_id, status FROM attendance;
```

ATTENDANCE_ID	EMPLOYEE_ID	STATUS
1	1	Present
2	2	Late
3	3	Present
4	4	Leave
5	5	Present

DELETE FROM attendance

WHERE status = 'Late';

COMMIT;

/* =====

SECTION 3: DQL - DATA QUERY LANGUAGE

SELECT queries

===== */

/* 1. View all employees */

```
SQL> SELECT *
FROM employees;
2
EMPLOYEE_ID FIRST_NAME
-----
LAST_NAME
-----
EMAIL
-----
PHONE HIRE_DATE SALARY DEPARTMENT_ID JOB_ID
-----
MANAGER_ID EMPLOYMENT_STATUS
-----
1 Aung
Aung
aung.aung@company.com
EMPLOYEE_ID FIRST_NAME
-----
LAST_NAME
-----
EMAIL
-----
PHONE HIRE_DATE SALARY DEPARTMENT_ID JOB_ID
-----
MANAGER_ID EMPLOYMENT_STATUS
-----
0911111111 10-JAN-22 2600000 3 5
Active
```

/* 2. Employee with department and job */

```
SQL> SELECT
      e.employee_id,
      e.first_name || ' ' || e.last_name AS employee_name,
      d.department_name,
      j.job_title,
      e.salary
FROM employees e
JOIN departments d
      ON e.department_id = d.department_id
JOIN jobs j
      ON e.job_id = j.job_id;
```

2	3	4	5	6	7	8	9	10	11
---	---	---	---	---	---	---	---	----	----

EMPLOYEE_ID

EMPLOYEE_NAME

DEPARTMENT_NAME

JOB_TITLE

SALARY

2

Su Su

Human Resources

EMPLOYEE_ID

EMPLOYEE_NAME

DEPARTMENT_NAME

JOB_TITLE

SALARY

HR Officer

800000

/* 3. Employees with salary greater than 1,000,000 */

```
SQL> SELECT employee_id, first_name, last_name, salary
FROM employees
WHERE salary > 1000000;
```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	SALARY
1	Aung	Aung	2600000
4	Hla	Hla	1300000

/* 4. Count employees by department */

```
SQL> SELECT
    d.department_name,
    COUNT(e.employee_id) AS total_employees
FROM departments d
LEFT JOIN employees e
    ON d.department_id = e.department_id
GROUP BY d.department_name
ORDER BY total_employees DESC;
```

DEPARTMENT_NAME	TOTAL_EMPLOYEES
IT	2
Sales	1
Human Resources	1
Finance	1

/* 5. Average salary by department */

```
SQL> SELECT
      d.department_name,
      AVG(e.salary) AS average_salary
FROM employees e
JOIN departments d
      ON e.department_id = d.department_id
GROUP BY d.department_name;
```

2	3	4	5	6	7
DEPARTMENT_NAME					

AVERAGE_SALARY					
----------------	--	--	--	--	--

Human Resources					
800000					

Finance					
950000					

IT					
1950000					

DEPARTMENT_NAME					
-----------------	--	--	--	--	--

AVERAGE_SALARY					
----------------	--	--	--	--	--

Sales					
700000					

/* 6. Highest salary employee */

```
SQL> SELECT *
FROM employees
WHERE salary = (
    SELECT MAX(salary)
    FROM employees
);
 2      3      4      5      6
EMPLOYEE_ID FIRST_NAME
-----
LAST_NAME
-----
EMAIL
-----
PHONE                HIRE_DATE      SALARY DEPARTMENT_ID  JOB_ID
-----
MANAGER_ID EMPLOYMENT_STATUS
-----
          1 Aung
Aung
aung.aung@company.com

EMPLOYEE_ID FIRST_NAME
-----
LAST_NAME
-----
EMAIL
-----
PHONE                HIRE_DATE      SALARY DEPARTMENT_ID  JOB_ID
-----
MANAGER_ID EMPLOYMENT_STATUS
-----
09111111111111111111 10-JAN-22      26000000          3          5
          Active
```

/* 7. Employee manager report */

```
SQL> SELECT
      e.first_name || ' ' || e.last_name AS employee_name,
      m.first_name || ' ' || m.last_name AS manager_name
FROM employees e
LEFT JOIN employees m
      ON e.manager_id = m.employee_id;
      2      3      4      5      6
EMPLOYEE_NAME
-----
MANAGER_NAME
-----
Su Su
Aung Aung

Kyaw Kyaw
Aung Aung

Hla Hla
Aung Aung

EMPLOYEE_NAME
-----
MANAGER_NAME
-----
Mya Mya
Aung Aung

Aung Aung
```

/* 8. Attendance report */

```
SQL> SELECT
  e.first_name || ' ' || e.last_name AS employee_name,
  a.attendance_date,
  a.check_in,
  a.check_out,
  a.status
FROM attendance a
JOIN employees e
  ON a.employee_id = e.employee_id;
```

2 3 4 5 6 7 8 9

EMPLOYEE_NAME

ATTENDANC

CHECK_IN

CHECK_OUT

STATUS

Aung Aung

01-MAY-26

01-MAY-26 08.55.00.000000 AM

EMPLOYEE_NAME

ATTENDANC

CHECK_IN

CHECK_OUT

STATUS

01-MAY-26 05.30.00.000000 PM

Present

/* 9. Payroll report */

```
SQL> SELECT
      e.first_name || ' ' || e.last_name AS employee_name,
      p.salary_month,
      p.basic_salary,
      p.allowance,
      p.deduction,
      p.net_salary
FROM payroll p
JOIN employees e
  ON p.employee_id = e.employee_id;
```

2 3 4 5 6 7 8 9 10
EMPLOYEE_NAME

SALARY_MONTH	BASIC_SALARY	ALLOWANCE	DEDUCTION	NET_SALARY
--------------	--------------	-----------	-----------	------------

Aung Aung May-2026	2500000	300000	100000	2700000
-----------------------	---------	--------	--------	---------

Su Su May-2026	800000	100000	30000	870000
-------------------	--------	--------	-------	--------

Kyaw Kyaw May-2026	950000	150000	50000	1050000
-----------------------	--------	--------	-------	---------

EMPLOYEE_NAME

SALARY_MONTH	BASIC_SALARY	ALLOWANCE	DEDUCTION	NET_SALARY
--------------	--------------	-----------	-----------	------------

Hla Hla May-2026	1200000	200000	80000	1320000
---------------------	---------	--------	-------	---------

Mya Mya May-2026	700000	80000	20000	760000
---------------------	--------	-------	-------	--------

/* 10. Employees hired in 2024 */

```
SQL> SELECT employee_id, first_name, last_name, hire_date
FROM employees
WHERE EXTRACT(YEAR FROM hire_date) = 2024;
```

2	3	
EMPLOYEE_ID	FIRST_NAME	
		LAST_NAME
		HIRE_DATE
4	Hla	
	Hla	01-FEB-24
5	Mya	
	Mya	05-MAY-24

/* =====
SECTION 4: DTL / TCL - TRANSACTION CONTROL LANGUAGE
COMMIT, ROLLBACK, SAVEPOINT
===== */

/* Transaction example */

SAVEPOINT before_salary_update;

```
UPDATE employees
SET salary = salary + 50000
WHERE department_id = 1;
```

/* Check updated data */

```
SQL> SELECT employee_id, first_name, salary
FROM employees
WHERE department_id = 1;
```

2	3	
EMPLOYEE_ID	FIRST_NAME	SALARY
2	Su	850000

/* Rollback only to savepoint */
ROLLBACK TO before_salary_update;

```
/* Final commit */  
COMMIT;  
/* Another transaction example */
```

```
SAVEPOINT before_delete;
```

```
DELETE FROM payroll  
WHERE salary_month = 'May-2026'  
AND employee_id = 5;
```

```
/* Cancel delete */  
ROLLBACK TO before_delete;
```

```
COMMIT;
```

```
/* =====  
SECTION 5: DCL - DATA CONTROL LANGUAGE  
User, Grant, Revoke  
Run by DBA/Admin user  
===== */
```

```
/* Create user */
```

```
CREATE USER hr_report_user IDENTIFIED BY Report##123;
```

```
/* Allow user to login */
```

```
GRANT CREATE SESSION TO hr_report_user;
```

```
/* Grant SELECT permission */
```

```
GRANT SELECT ON departments TO hr_report_user;  
GRANT SELECT ON employees TO hr_report_user;  
GRANT SELECT ON jobs TO hr_report_user;  
GRANT SELECT ON attendance TO hr_report_user;
```

```
GRANT SELECT ON payroll TO hr_report_user;
```

```
/* Grant INSERT permission only on attendance */
```

```
GRANT INSERT ON attendance TO hr_report_user;
```

```
/* Revoke INSERT permission */
```

```
REVOKE INSERT ON attendance FROM hr_report_user;
```

```
/* Create role */
```

```
CREATE ROLE hr_readonly_role;
```

```
/* Grant permissions to role */
```

```
GRANT SELECT ON departments TO hr_readonly_role;
```

```
GRANT SELECT ON employees TO hr_readonly_role;
```

```
GRANT SELECT ON jobs TO hr_readonly_role;
```

```
GRANT SELECT ON attendance TO hr_readonly_role;
```

```
GRANT SELECT ON payroll TO hr_readonly_role;
```

```
/* Assign role to user */
```

```
GRANT hr_readonly_role TO hr_report_user;
```

```
/* Drop user example */
```

```
-- DROP USER hr_report_user CASCADE;
```

```
/* =====  
SECTION 6: CLEANUP SCRIPT  
Use only if you want to delete all objects  
===== */
```

```
/*  
DROP TABLE payroll CASCADE CONSTRAINTS;  
DROP TABLE attendance CASCADE CONSTRAINTS;  
DROP TABLE employees CASCADE CONSTRAINTS;  
DROP TABLE jobs CASCADE CONSTRAINTS;  
DROP TABLE departments CASCADE CONSTRAINTS;  
  
DROP SEQUENCE seq_department;  
DROP SEQUENCE seq_job;  
DROP SEQUENCE seq_employee;  
DROP SEQUENCE seq_attendance;  
DROP SEQUENCE seq_payroll;  
  
DROP ROLE hr_readonly_role;  
DROP USER hr_report_user CASCADE;  
*/
```